STORAGE WARS

THE MECHANICAL HDD

Section 01 / 04

Think of it like a Record Player

Magnetic platters spin at 7,200 RPM while a tiny arm reads & writes data.

THE PROBLEM

Mechanical = Slow & Fragile. Dropping a laptop can cause a HEAD CRASH — permanent data loss.

Avg. Read Speed: ~100–150 MB/s

HDD = Hard Disk Drive | Magnetic Storage | Moving Parts | Mechanical Arm

STORAGE WARS

FORM FACTORS: SATA vs. NVMe

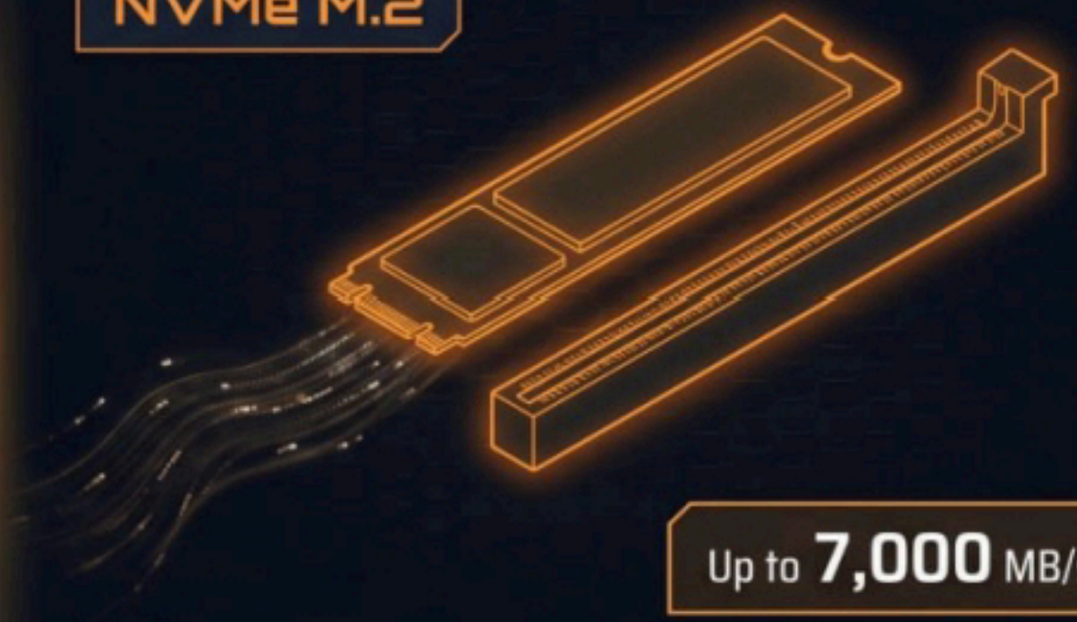
Section 03 / 04

SATA SSD

Up to **550** MB/s

Uses same cables as HDDs. Older interface.
Speed is limited by the SATA bottleneck.

NVMe M.2



Up to **7,000** MB/s

Plugs directly into motherboard. No cables.
Up to 10× faster than SATA.

⚠ **THE PROBLEM:** Speed Creates Heat — High-performance NVMe drives risk THERMAL THROTTLING, slowing down to protect circuits.

STORAGE WARS

SUMMARY & COMMON FAILURES

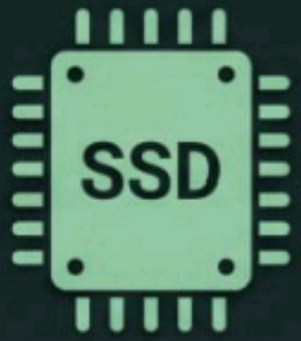
Section 04 / 04

HDD

✓ Best for:
Large Storage (8TB+)
— Low Cost

✕ Fails Due To:
Physical wear
Clicking noises
Vibration sensitivity
Head Crash risk

SSD



✓ Best for:
Speed, Silence
Portability

✕ Fails Due To:
Electronic wear on
NAND cells
Controller malfunctions
Write endurance limits

THE GOLDEN RULE

ALL DRIVES EVENTUALLY DIE.

The best solution isn't a repair
— it's a **BACKUP.**

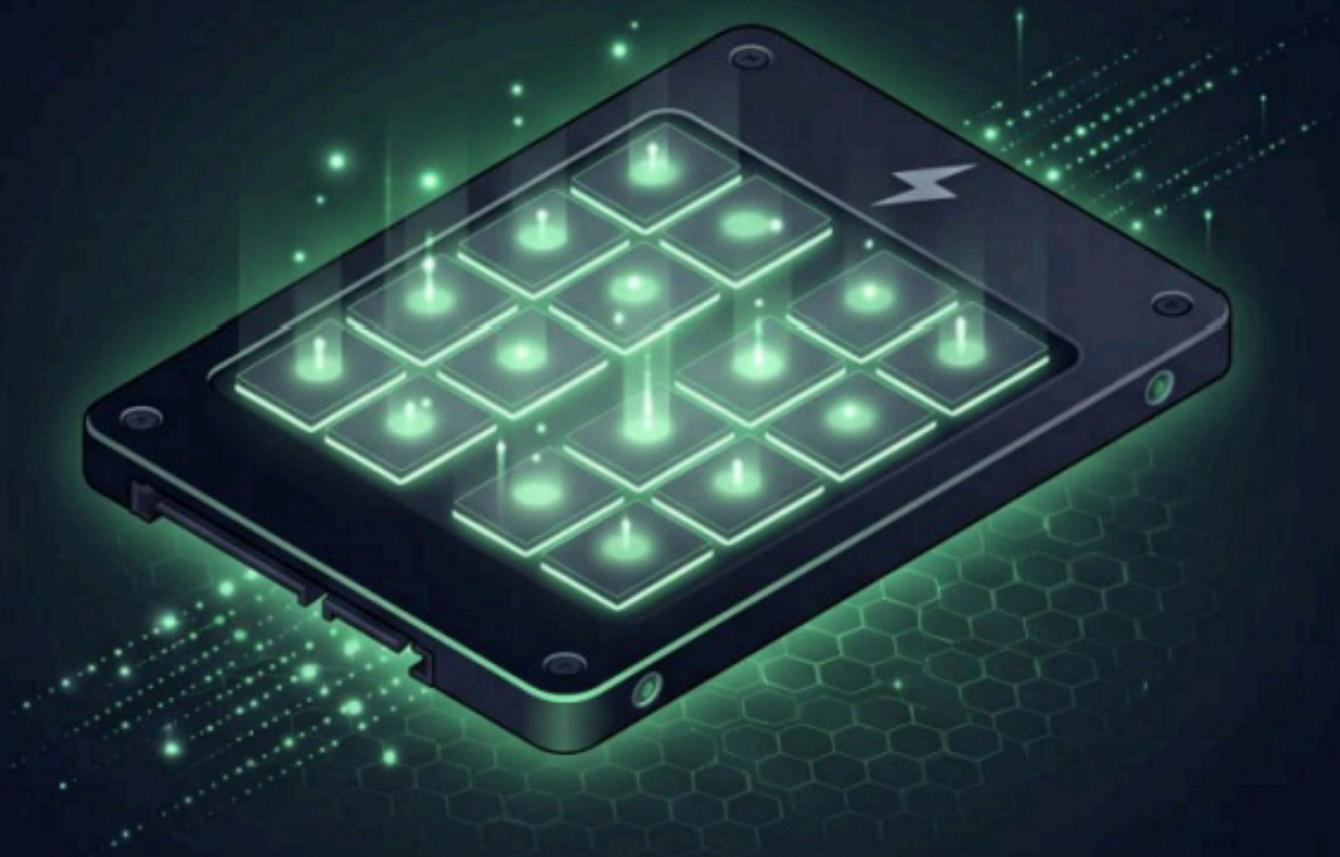
CTA: 3-2-1 Backup Rule:
3 copies, 2 media types, 1 offsite.

Drive Type	Speed	Durability	Cost	Best Use
HDD	100 Mbp/s	Durability	$I Cost	Large Storage (8TB+)
SATA SSD	300 Mbp/s	Durability	$5 Cost	Large Storage
NVMe SSD	1000 Mbp/s	Vanidity	$20 Cost	Best for Low Cost

STORAGE WARS

THE SOLID STATE REVOLUTION

Section 02 / 04



NAND Flash Memory

No moving parts. Same tech as USB sticks — but dramatically faster and more reliable.

NEARLY INSTANTANEOUS

No mechanical arm to move = data access in microseconds.

THE PROBLEM: Write Endurance

Every save wears the cells slightly. Heavy servers face a constant race against time.

Avg. Read Speed: ~500 MB/s (SATA SSD)

SSD = Solid State Drive | NAND Flash | No Moving Parts | Electronic Storage